8.Perform basic Image Handling and processing operations on the image is to read an image in python and Dilate an Image using Dilate function

code;

```
import cv2
import matplotlib.pyplot as plt

# Read the image
img = cv2.imread(
    r"C:\Users\ROHIT\Pictures\Instagram\IMG_20230206_213414_457.jpg"
)

if img is None:
    print("Image not found!")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Create kernel
    kernel = cv2.getStructuringElement(
        cv2.MORPH_RECT, (5, 5)
    )

    # Apply dilation
    dilated = cv2.dilate(gray, kernel, iterations=1)

    # Display images
    plt.subplot(1, 2, 1)
    plt.imshow(gray, cmap='gray')
    plt.title("Original Image")
```

```
plt.axis("off")


plt.subplot(1, 2, 2)
plt.imshow(dilated, cmap='gray')
plt.title("Dilated Image")
plt.axis("off")
plt.show()
```

output

